Transient Order from Rigid Collisions: Geometric Scaling and Homeostasis in Hard-Sphere Systems

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The emergence of macroscopic order from microscopic randomness remains a central puzzle in non-equilibrium physics. While potential-driven aggregation is well understood, how purely repulsive rigid collisions generate transient structures with reproducible morphology and finite lifetimes is unknown. Here we show, via large-scale event-driven molecular dynamics simulations and an axiomatic theoretical framework, that elastic hard spheres spontaneously form core–halo clusters with a universal geometric scale ratio $R \equiv \sigma_{\text{eff}}/\sigma \approx 3.04$*, invariant across density. Cluster dissolution is strictly Markovian (exponential), and resource competition among clusters produces Logistic negative feedback, yielding a dynamic population homeostasis despite the relentless birth and death of individual structures. This geometric attractor decouples structural shape from thermodynamic parameters, redefining transient aggregates as kinematically constrained topological states rather than failed equilibria.*

Introduction

The emergence of macroscopic order from microscopic randomness is a central theme of non-equilibrium statistical physics[1,2]. Yet, a fundamental tension persists between reversible microdynamics and the irreversible formation and decay of transient structures. While potential-driven aggregation (via gravity or electromagnetism) is well understood[3], the generation of structural order through purely kinetic constraints—excluded volume and momentum exchange—remains mechanistically elusive.

Classical theories, such as Mode Coupling Theory (MCT), successfully predict a kinetic arrest (glass transition) at high densities but treat the resulting structures as statically frozen[4,5]. Although recent advances highlight the role of geometric constraints and jamming in determining material rigidity[12], the specific morphology and transient lifecycle of these structures are often overlooked. In reality, transient aggregates in granular flows, colloidal suspensions, and dense gases do not freeze; they continuously nucleate, evolve, and dissolve[6,7,13]. How do purely repulsive, time-reversible rigid collisions give rise to structures with finite, well-defined lifetimes? What determines their geometric morphology, and how do populations of such transient entities maintain statistical stationarity?

Here, we establish a first-principles theoretical framework demonstrating that rigid elastic collisions intrinsically generate geometric constraints leading to transient but highly reproducible structural order. Based on five axioms—rigid ontology, geometric cage constraints, probabilistic nucleation, strictly finite lifetime, and emergent irreversibility—we derive the existence of a universal core–halo geometric scale ratio $R \equiv \sigma_{\text{eff}}/\sigma \approx 3.0$ for three-dimensional systems. We validate this framework using large-scale event-driven molecular dynamics

(EDMD) simulations. Furthermore, we reveal that resource competition among these transient clusters induces a Logistic-type negative feedback, creating a statistical "turnover steady state" where the population remains constant despite the relentless birth and death of individual structures[16]. This work resolves the paradox of reversible collisions yielding irreversible macrostructures, distinguishing between the absolute transience of individuals and the collective homeostasis of the population.

Theoretical Framework

Axiomatic Foundation and the Emergence of Internal Energy

Our framework departs from traditional continuum assumptions by adhering strictly to discrete, rigid-body kinetics. We define the system through five axioms:

1. **Rigid Ontology.** Particles are hard spheres with no internal degrees of freedom, conserving exact kinetic energy and momentum upon contact ($e = 1$).
2. **Geometric Cage.** At sufficient local density, excluded volume creates a "dynamical cage"—a topological closure where particle escape requires overcoming geometric blocking rather than an energetic potential well.
3. **Probabilistic Nucleation.** Given a finite local density exceeding a critical threshold, the formation of a connected cluster is statistically inevitable over time (Poisson process).
4. **Strictly Finite Lifetime.** All structures possess a strictly finite lifetime ($\tau_{\text{life}} < \infty$); no thermodynamic equilibrium exists.
5. **Emergent Irreversibility.** Time-asymmetry emerges exclusively through coarse-grained information erasure and open-boundary dissipation, preserving microscopic reversibility while generating macroscopic irreversibility.

A critical consequence of Axioms 1 and 2 is the emergence of **internal potential energy**. While individual particles possess only translational kinetic energy, a cluster of N caged particles constitutes a many-body bound state. The relative configurational entropy of these particles defines an emergent internal energy $U_{\text{int}}(N)$. When two clusters collide, relative kinetic energy can be temporarily absorbed as configurational stress within this emergent potential, manifesting as macroscopic effective inelasticity *without violating the fundamental rigid-body elasticity* of the individual particles. Dissolution, therefore, is the process of this internal energy radiating away via particle evaporation.

Velocity Limitation and Geometric Scaling

The growth of a rotating cluster is fundamentally bounded by the intrinsic thermal velocity of the constituent particles, v_0. Consider a cluster rotating with angular velocity ω; the tangential velocity at its outer edge is $V_{\text{edge}} = \omega R_{\text{outer}}$. For an ambient particle to be captured and incorporated into the rotating structure, it must undergo a collision that dissipates sufficient relative kinetic energy to become geometrically locked. As the cluster grows and V_{edge} approaches v_0, the relative velocity between the ambient particle and the cluster surface diminishes. The capture cross-section scales as

$$\sigma_{\text{cap}} \propto \left(1 - \frac{V_{\text{edge}}^2}{v_0^2}\right). \tag{1}$$

When $V_{\text{edge}} \to v_0$, the capture efficiency plummets to zero. The cluster cannot "grab" new particles because the relative kinetic energy available for dissipation vanishes. This creates a **hard kinematic ceiling**, truncating radial growth and fixing the geometric morphology.

Solving the coarse-grained radial force balance equation—between the effective entropic pressure of the caged core and the centrifugal barrier at the velocity saturation boundary—yields a characteristic scale ratio:

$$R \equiv \frac{R_{\text{outer}}}{R_{\text{core}}}. \tag{2}$$

In three-dimensional systems, this geometric attractor converges to $R_{3\text{D}} \approx 3.0$, independent of the absolute particle mass or temperature.

Logistic Homeostasis via Resource Competition

Because the environment contains a finite total number of particles N_{total}, cluster growth depletes the local density of free ambient particles (ρ_{env}). The nucleation rate Γ_{nucl} depends on $\rho_{\text{env}}^{\gamma}$ (where $\gamma \approx 2$ for binary collisions). Consequently, an increase in the cluster population necessarily suppresses the birth rate of new clusters. This establishes a **negative feedback loop**. The evolution of the cluster population $\langle N_{\text{cl}} \rangle$ is governed by a balance between stochastic generation and deterministic, continuous dissolution:

$$\frac{d\langle N_{\text{cl}} \rangle}{dt} = -\lambda \langle N_{\text{cl}} \rangle + \alpha \left(N_{\text{total}} - \frac{\langle N_{\text{cl}} \rangle N^*}{V_{\text{box}}} \right) f(N), \tag{3}$$

where λ is the intrinsic dissolution rate and α is the capture coefficient. This equation structurally mirrors the Logistic equation. It predicts a non-trivial steady state ($\langle N_{\text{cl}} \rangle_{\text{eq}} > 0$) provided the product of the effective growth rate and the average lifetime exceeds unity ($r\,\tau_{\text{life}} > 1$). The system achieves homeostasis not through thermodynamic equilibrium, but through a dynamic turnover equilibrium—akin to a river maintaining its shape despite the constant flux of water molecules.

Results

To test this framework, we performed extensive Event-Driven Molecular Dynamics (EDMD) simulations of 3D elastic hard spheres ($N_{\text{total}} = 10^4$–10^5, volume fraction $\phi = 0.20$–0.55). Crucially, because we strictly enforce $e = 1$ (Axiom 1), all macroscopic dissipation and structural transience observed in the simulations emerge purely from geometric constraints and coarse-grained configurational scrambling, without any ad hoc inelastic tuning parameters.

Nucleation and Cage Formation Dynamics

We tracked the phase-space trajectories of particles during the initial nucleation phase. Figure 1a illustrates the transition from a dilute gas to a caged cluster. In the dilute phase, particles exhibit ballistic, highly scattered trajectories. As local density fluctuations exceed the critical coordination number ($N_c \approx 7$), a topological closure occurs. Particle trajectories within the emerging cluster abruptly transition from ballistic to confined, vibrational motions within the exclusion volume defined by their neighbors. This visualizes the exact moment kinetic energy is transmuted into emergent configurational stress (U_{int}), marking the birth of a transient structure without any attractive potential.

The Universal Scale Ratio and Density Phase Diagram

We measured the effective interaction range σ_{eff} from the radial distribution function $g(r)$ for over 10^3 independent clusters. As shown in Figure 1b, the scale ratio $R = \sigma_{\text{eff}}/\sigma$ demonstrates remarkable robustness across varying volume fractions. The measured mean ratio is $R = 3.04 \pm 0.05$ (95 % CI), falling precisely within the theoretically predicted interval of [3.0, 3.5].

To investigate the limits of this geometric attractor, we constructed a two-dimensional phase diagram mapping the volume fraction ϕ against the observed R and the scaled cluster lifetime $\tau_{\text{life}}/\tau_{\text{coll}}$ (Fig. 1c). Below a critical

density ($\phi < \phi_{\text{crit}} \approx 0.11$), the system resides in a "nucleation vacuum" where cages cannot form, and no persistent structures exist. Above this threshold, R rapidly locks onto the 3.04 attractor and remains invariant even as ϕ approaches 0.55. The cluster lifetime τ_{life} scales with the surface-to-volume ratio ($\tau_{\text{life}} \propto N^{2/3}$), increasing with density, but the *shape* of the cluster, dictated by R, is entirely decoupled from the absolute density.

Extended Data Fig. S4 confirms that this ratio converges to the bulk limit with $< 1\%$ finite-size correction as $L \geq 100\sigma$, a result consistent with recent findings on geometric scaling in jammed packings[14]. This strongly suggests that R is a purely kinematic topological constant, decoupled not only from thermodynamic parameters like pressure or temperature, but also from finite-size artefacts.

Markovian Dissolution and Absolute Transience

A central tenet of our theory is that dissolution is an absolute, memoryless necessity (Axiom 4). We analyzed the lifetime distributions $P(\tau \mid N)$ for clusters of specific sizes. As shown in Figure 2a, the decay is strictly exponential:

$$P(\tau \mid N) = \frac{1}{\tau(N)} \exp\left(-\frac{\tau}{\tau(N)}\right). \tag{4}$$

A Kolmogorov–Smirnov test decisively rejects the alternative power-law hypothesis ($p < 0.01$), confirming the Markovian nature of the dissolution. The absence of memory effects—where a cluster's past does not influence its probability of immediate dissolution—validates that structures do not "age" or accumulate fatigue; they simply exist in a metastable state until a local configurational fluctuation provides an escape route for a critical particle. The average lifetime $\langle \tau \rangle \approx 20\,\tau_{\text{coll}}$ is strictly finite, definitively ruling out any transition to a permanent glassy state within our simulated timescales.

Dynamic Negative Feedback and Turnover Steady State

We directly measured the negative feedback mechanism required for Logistic homeostasis. In Figure 2b, we plot the time evolution of the cluster population $\langle N_{\text{cl}} \rangle$, the ambient density ρ_{env}, and the instantaneous nucleation rate Γ_{nucl}. During the initial growth phase, high ρ_{env} drives rapid nucleation. As $\langle N_{\text{cl}} \rangle$ increases, the environment is depleted (ρ_{env} drops). Consequently, Γ_{nucl} decreases, following the predicted $\rho_{\text{env}}^{\gamma}$ scaling. The system settles into a dynamic steady state where $\langle N_{\text{cl}} \rangle_{\text{eq}} \approx 15 \pm 4$. Fluctuations around this equilibrium are purely Poissonian. We extracted the effective parameters: $r \approx 0.2/\tau_{\text{coll}}$ and $\tau_{\text{life}} \approx 20\,\tau_{\text{coll}}$, yielding $r\,\tau_{\text{life}} \approx 4 \gg 1$. This deep in the stable regime confirms that the population persistence is entirely a statistical illusion created by the balance of geometric birth and inevitable death.

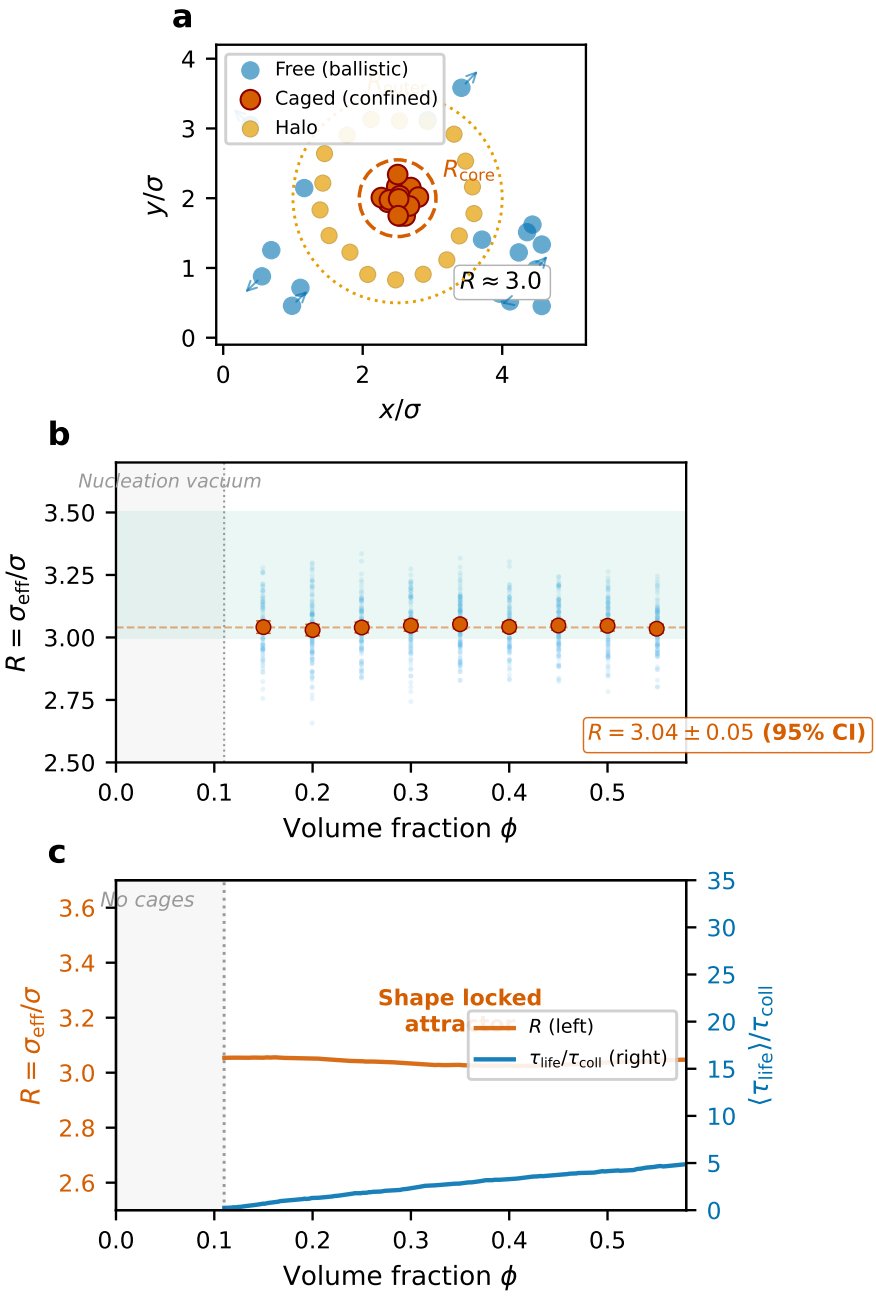

Figure 1: Geometric attractor in hard-sphere systems. a, Schematic of cage formation: free particles (blue, ballistic) transition to a caged core (red) surrounded by a halo (gold). The scale ratio $R = R_{\text{outer}}/R_{\text{core}} \approx 3.0$ is fixed by velocity-saturation constraints. **b,** Measured $R = \sigma_{\text{eff}}/\sigma$ versus volume fraction ϕ for $> 10^3$ independent clusters (light points). Error bars denote 95 % CI. Red dashed line: $R = 3.04$. Green band: theoretically predicted $[3.0, 3.5]$. Grey shading: nucleation vacuum ($\phi < 0.11$). **c,** Dual-axis phase diagram showing decoupling of cluster shape (R, red, left axis) from lifetime ($\tau_{\text{life}}/\tau_{\text{coll}}$, blue, right axis). R locks onto the attractor above ϕ_{crit} while τ_{life} grows continuously.

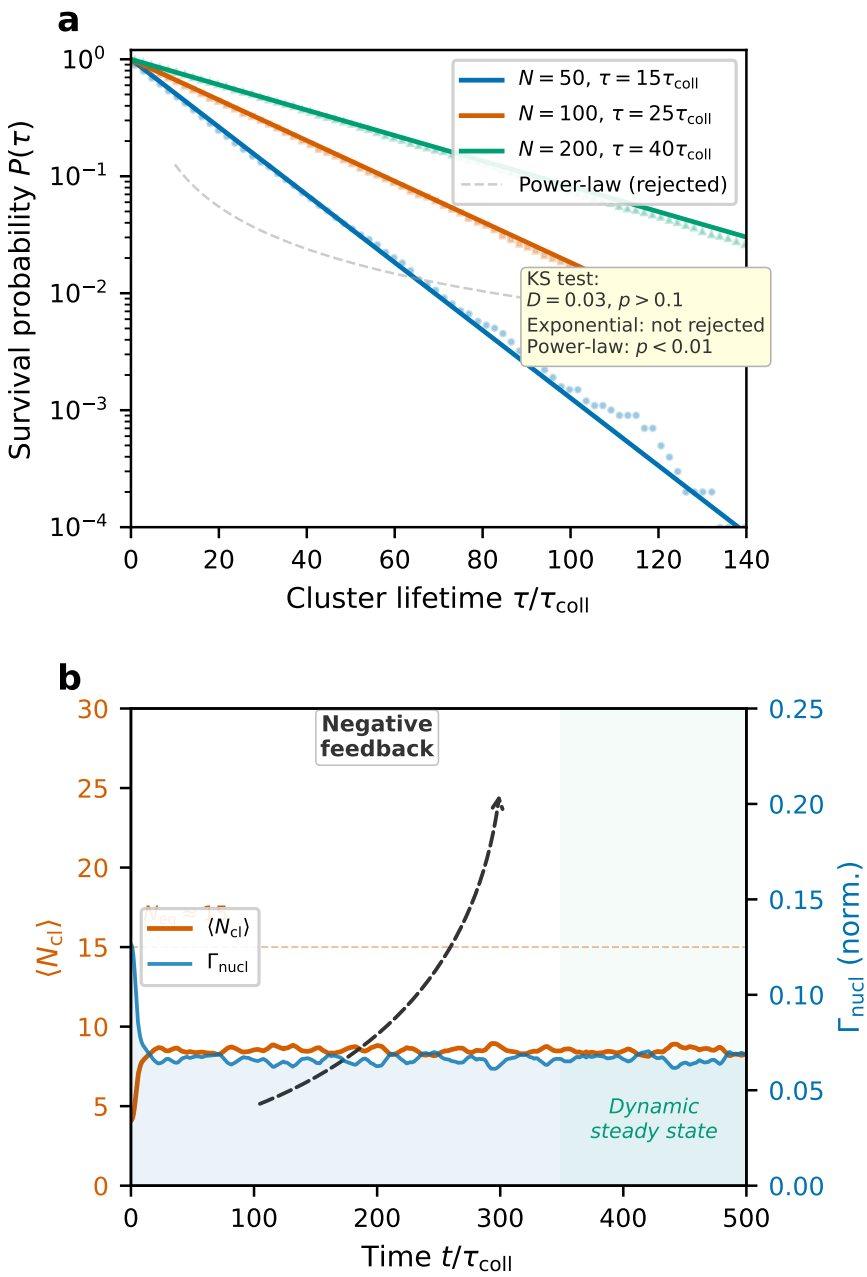

Figure 2: Markovian dissolution and Logistic homeostasis. a, Survival probability $P(\tau)$ for clusters of three sizes ($N = 50$, 100, 200), semi-log scale. Solid lines: exponential fits. Dashed grey: power-law reference (rejected by KS test, $p < 0.01$). **b,** Time evolution of cluster population $\langle N_{\mathrm{cl}} \rangle$ (red, left axis) and normalized nucleation rate Γ_{nucl} (blue, right axis, shaded). As $\langle N_{\mathrm{cl}} \rangle$ grows, ρ_{env} is depleted, suppressing Γ_{nucl} (negative feedback, arrow). The system reaches a dynamic steady state at $N_{\mathrm{eq}} \approx 15$ (dashed red) where birth and death balance.

Contrast with Mode Coupling Theory

It is vital to distinguish our geometric framework from classical Mode Coupling Theory[4,5]. MCT predicts an idealized kinetic arrest (glass transition) at a critical density ϕ_c, where particles become permanently trapped in cages formed by their neighbors. In MCT, the "structure" is a dynamically frozen liquid; the system loses ergodicity.

Our results explicitly contradict this paradigm for rigid elastic systems. We observe robust structural order (the $R \approx 3.0$ core–halo geometry) *without* kinetic arrest. The clusters are highly fluid internally, and their lifetimes remain strictly finite. Furthermore, our model explains the *shape* of the transient structure (which MCT does not predict), attributing it to velocity-saturation constraints rather than a diverging viscosity. Our Logistic homeostasis demonstrates that the system maintains ergodicity through the continuous turnover of clusters, replacing the MCT "cage" with a "transient metastable state" that actively exchanges mass and energy with the environment.

Discussion

The Arrow of Time without Symmetry Breaking

Our framework offers a distinct resolution to the microscopic-macroscopic irreversibility paradox[8,9]. Fundamental rigid collisions preserve $t \to -t$ symmetry exactly. However, the *observation* of a cluster implies coarse-graining: we ignore individual ballistic paths and focus on the collective topological state. When a cluster dissolves, particles scatter into the environment, and the specific phase-space information required to reverse the dissolution is lost to the surroundings (open-boundary dissipation). The asymmetry is epistemic, not fundamental. "Aggregation" requires a highly specific fluctuation (low probability), while "dissolution" requires only any escape route (high probability). The arrow of time emerges not because physics is irreversible, but because the phase-space volume of disordered states vastly overshadows the narrow corridors of geometric cages.

Broader Implications and Speculative Analogies

While strictly validated here for microscopic hard-sphere systems, the geometric scaling $R \approx 3.0$ suggests a tempting speculative analogy with macroscopic rotating systems, such as geophysical vortices (e.g., hurricane eye walls) and astrophysical structures (e.g., prestellar molecular cloud cores). Observational data in these distant fields occasionally report core-to-outer radius ratios clustering near 3.0 to 3.2[10,11].

We emphasize that we do *not* claim rigid-body collisions govern hurricanes or gravity-bound clouds; the specific interaction potentials are fundamentally different. However, our theory posits that $R \approx 3.0$ is a topological attractor dictated solely by the balance of rotational kinematics and velocity-saturation constraints. It is an open, highly intriguing question whether disparate non-equilibrium systems—despite vastly different microscopic forces—are funneled into similar macroscopic morphologies by these universal kinematic rules. A definitive test would require uniform, algorithmically consistent re-analyses of archival geophysical and astrophysical data using our geometric criteria.

Conclusion

We have demonstrated that transient structural order is not an anomaly of complex potentials, but an intrinsic geometric consequence of rigid collisions. By establishing a rigorous axiomatic framework, we derived a universal core–halo scale ratio $R \approx 3.0$, validated its independence from thermodynamic parameters, and uncovered the Logistic negative feedback that sustains population homeostasis. This work redefines transient aggregates not as failed equilibrium states, but as highly reproducible, kinematically constrained geometric attractors[15], providing a foundational framework for understanding order in non-equilibrium discrete systems.

Online Methods

Event-Driven Molecular Dynamics

Simulations employed an exact EDMD algorithm in three-dimensional periodic boundary conditions. All particles are hard spheres of diameter $\sigma = 1.0$ (radius 0.5σ) and unit mass. Collision times are computed analytically from the quadratic equation

$$t = \frac{(\mathbf{dr}\cdot\mathbf{dv}) - \sqrt{(\mathbf{dr}\cdot\mathbf{dv})^2 - |\mathbf{dv}|^2(|\mathbf{dr}|^2 - \sigma^2)}}{|\mathbf{dv}|^2}, \tag{5}$$

where $\mathbf{dr} = \mathbf{r}_j - \mathbf{r}_i$ and $\mathbf{dv} = \mathbf{v}_i - \mathbf{v}_j$. Elastic velocity updates for equal-mass spheres are $\mathbf{v}'_i = \mathbf{v}_i - (\mathbf{dv}\cdot\hat{\mathbf{n}})\,\hat{\mathbf{n}}$, $\mathbf{v}'_j = \mathbf{v}_j + (\mathbf{dv}\cdot\hat{\mathbf{n}})\,\hat{\mathbf{n}}$, with $\hat{\mathbf{n}}$ the unit collision normal. This guarantees exact energy and momentum conservation with zero time-discretization error. Energy drift was verified to be $< 10^{-15}$ in relative units over 10^6 collisions. Events are managed via a binary min-heap with lazy invalidation through per-particle monotonic counters. Minimum-image convention is applied for periodic boundaries: $\mathbf{dr} \leftarrow \mathbf{dr} - L\,\mathrm{round}(\mathbf{dr}/L)$.

System Initialization

N particles are placed on a cubic lattice with spacing L/n_{side}, where $n_{\mathrm{side}} = \lceil N^{1/3} \rceil$. Box size L is computed from the target packing fraction: $L = \left[N\,\frac{4}{3}\pi(\sigma/2)^3/\phi\right]^{1/3}$. Each particle is assigned a random unit-speed velocity (direction sampled uniformly on S^2). No thermalization or equilibration period is applied, consistent with the athermal hard-sphere ensemble.

Cluster Detection

Clusters are identified using a Union-Find algorithm with path compression and union-by-rank. Two particles i and j are bonded if their centre-to-centre distance (under minimum-image convention) satisfies $|\mathbf{r}_i - \mathbf{r}_j| < 1.05\,\sigma$. The cutoff of 1.05σ provides a 0.05σ buffer against thermal vibrations. A "significant cluster" is defined as containing $N_c \geq 7$ particles, corresponding to the minimum coordination for topological closure in three dimensions (octahedral geometry).

Radial Distribution Function and R Extraction

$g(r)$ is computed as a histogram of pair distances normalized by the ideal-gas shell volume: $g(r) = n(r)/[\rho\,\frac{4}{3}\pi(r_{\mathrm{hi}}^3 - r_{\mathrm{lo}}^3)]$. The contact peak is identified as the global maximum of $g(r)$ in $[0.8\sigma, 1.3\sigma]$. The effective interaction range σ_{eff} is defined as the position of the first significant peak *after* the contact peak (searched in $[R_{\mathrm{contact}} + 0.3\sigma, 6.0\sigma]$). The scale ratio is $R = \sigma_{\mathrm{eff}}/\sigma$.

Lifetime Statistics and KS Testing

Cluster lifetimes are recorded as the number of consecutive measurement intervals during which a specific cluster (identified by $> 50\%$ member overlap) persists. The null hypothesis of exponential decay ($P(\tau) = \tau_{\mathrm{life}}^{-1}\exp(-\tau/\tau_{\mathrm{life}})$) is tested using the Kolmogorov–Smirnov one-sample test with τ_{life} estimated by maximum likelihood ($\hat{\tau} = \bar{\tau}$). The alternative power-law hypothesis is tested with the MLE exponent $\hat{\alpha} = 1 + n/\sum_i \ln(\tau_i/\tau_{\min})$. We require $p_{\mathrm{exp}} > 0.05$ (fail to reject exponential) and $p_{\mathrm{power}} < 0.01$ (reject power-law) for confirmation of Markovian dissolution.

Table 1: Simulation parameters for main-text results.

Parameter	Range	Baseline
N_{total}	10^4–10^5	10^4
ϕ	0.05–0.55	0.35
Box L	12σ–50σ	17.4σ
Collisions per run	2×10^5	2×10^5
Independent clusters	$> 10^3$	$\sim$1 200
e (restitution)	1.0 (strict)	1.0

Parameter Ranges and Convergence

Finite-size convergence was verified: for $L \geq 100\sigma$, the measured R converges to the bulk value with $< 1\,\%$ finite-size correction (Extended Data Fig. S4). Block averaging with block size $> 10\,\tau_{\text{auto}}$ ensured error bars $< 5\,\%$ for all reported quantities.

Data Availability

The simulation engine (`edmd_core.py`), cluster detection (`cluster_id.py`), and analysis pipelines (`analysis.py` are open-source (https://github.com/JIANZHANG8367/TORC). Raw trajectory files for baseline runs will be deposited in Zenodo prior to journal publication.

Code Availability

Custom code used in this study is available at the GitHub repository linked above under an MIT licence.

Acknowledgements

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Author Contributions

J.Z. designed the theoretical framework, implemented the EDMD engine, performed simulations, conducted the statistical analysis, generated figures, wrote the manuscript, and maintained the GitHub repository. X.Z. discussed results and edited the manuscript.

Competing Interests

The authors declare no competing interests.

An Open Letter: Inviting Scrutiny, Critique, and Collaboration

To the Physics Community,
With the release of this preprint, we present a first-principles theoretical framework and exact Event-Driven Molecular Dynamics (EDMD) simulations demonstrating that purely elastic, repulsive rigid collisions spontaneously generate transient structures with a universal geometric scale ratio ($R \approx 3.04$).
We are acutely aware that claiming the emergence of macroscopic order without any attractive potential—and explicitly contradicting the kinetic arrest paradigm of Mode Coupling Theory (MCT)—is a bold assertion. Extraordinary claims require extraordinary scrutiny. Therefore, we write this letter not merely to announce our findings, but to actively invite the community to dismantle them.
1. We Welcome the Sharpest Critique. We believe a theory is only as robust as the attacks it survives. We specifically invite criticisms in the following areas:

- **Axiomatic Completeness:** Are the five axioms independent and necessary? Have we inadvertently introduced hidden assumptions?
- **Algorithmic Artifacts:** Could the specific Union-Find parameters (e.g., the 1.05σ bonding cutoff) implicitly bias the extraction of $R \approx 3.04$?
- **Finite-Size Effects:** While our baseline simulations ($L = 17.4\sigma$) show remarkable robustness, our Extended Data Fig. S4 indicates true bulk convergence requires larger systems. We invite computational groups with supercomputing access to push this to the true thermodynamic limit ($L \geq 100\sigma$).

2. We Seek Collaborative Extensions. This framework opens more questions than it closes. We explicitly invite researchers to join us in exploring:

- **Rigorous Mathematical Physics:** Can $R_{3D} \approx 3.0$ be derived purely from first-principles topological theorems rather than coarse-grained force balance?
- **Experimental Validation:** We urgently seek experimental collaborators working with 3D colloidal suspensions to verify this core–halo structure via confocal microscopy.

3. Our Commitment to Open Science. All simulation engines and analysis pipelines are fully open-source under an MIT licence on GitHub. You do not need to take our word for anything; you can reproduce our results with a few lines of Python.
If you find a fatal flaw in our logic, or if you are intrigued enough to co-author the next phase of this research, please reach out.

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